

AWS 3-Tier Web Application Deployment

By - Vishal Konde

Project Goal

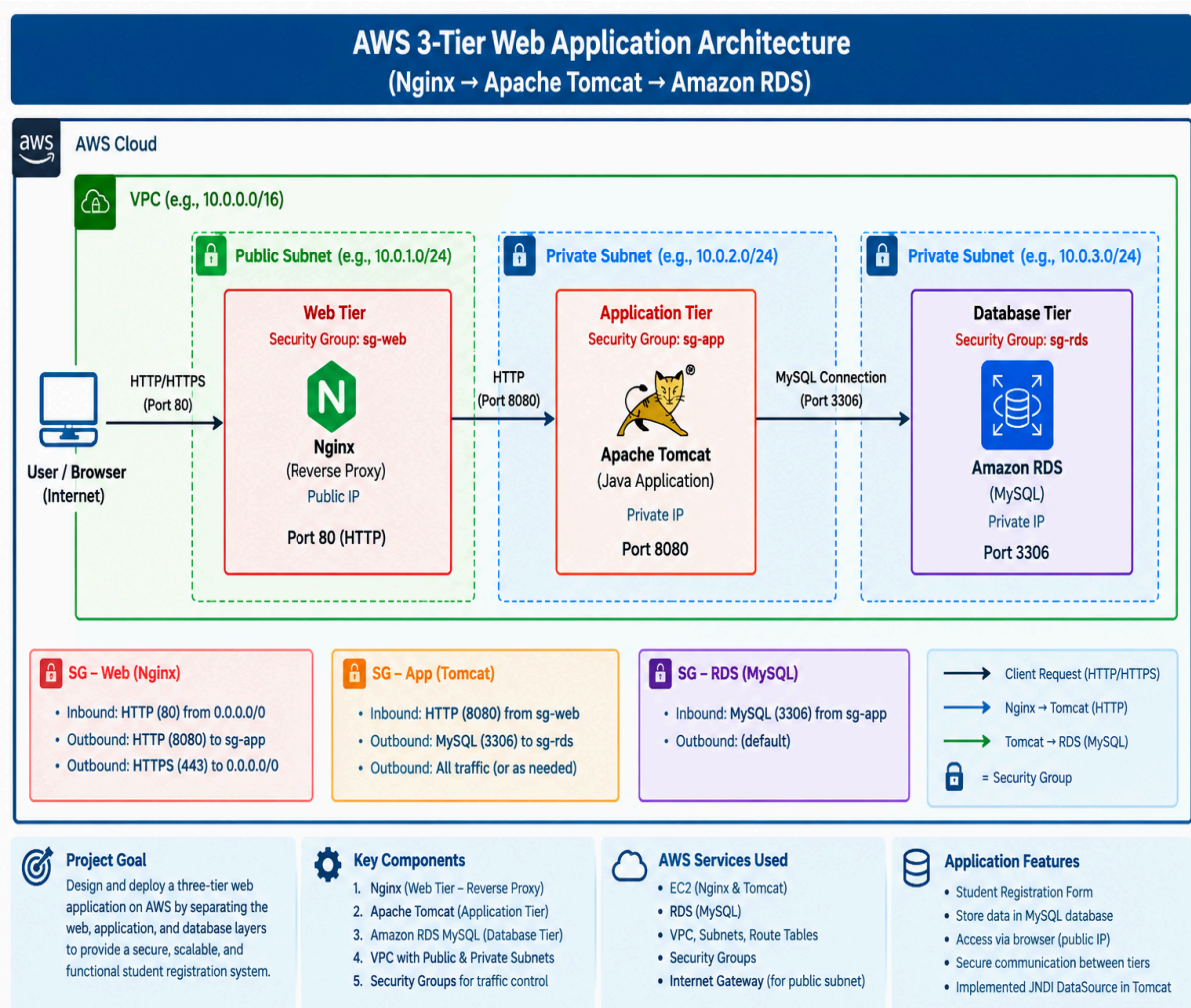
The goal of this project is to design and deploy a three-tier web application architecture on AWS by separating the web, application, and database layers. The project demonstrates how different AWS resources and application components communicate securely to provide a functional and scalable web application.

Project Objectives

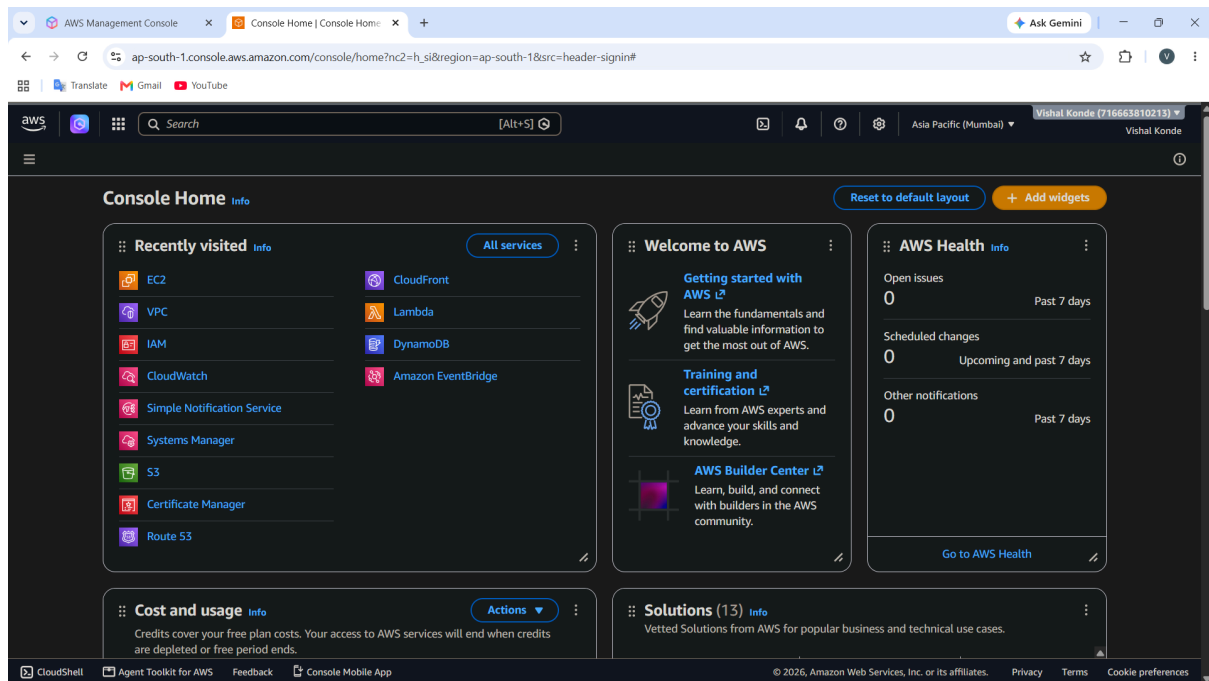
1. Deploy a Java-based Student Registration Application on AWS.
2. Configure Nginx as a reverse proxy to forward client requests to the application server.
3. Deploy the Java student.war application on Apache Tomcat.
4. Configure communication between the Nginx server and Tomcat application server.
5. Create an Amazon RDS MySQL database for persistent student records.
6. Connect the Java application to RDS using a **MySQL JDBC connector and JNDI DataSource**.

7. Configure Security Groups to control communication between the three tiers.
8. Store student registration details such as name, address, age, qualification, percentage, and passing year in the database.
9. Troubleshoot application connectivity across Nginx → Tomcat → RDS.
10. Demonstrate a practical implementation of a three-tier architecture in AWS.

Architecture



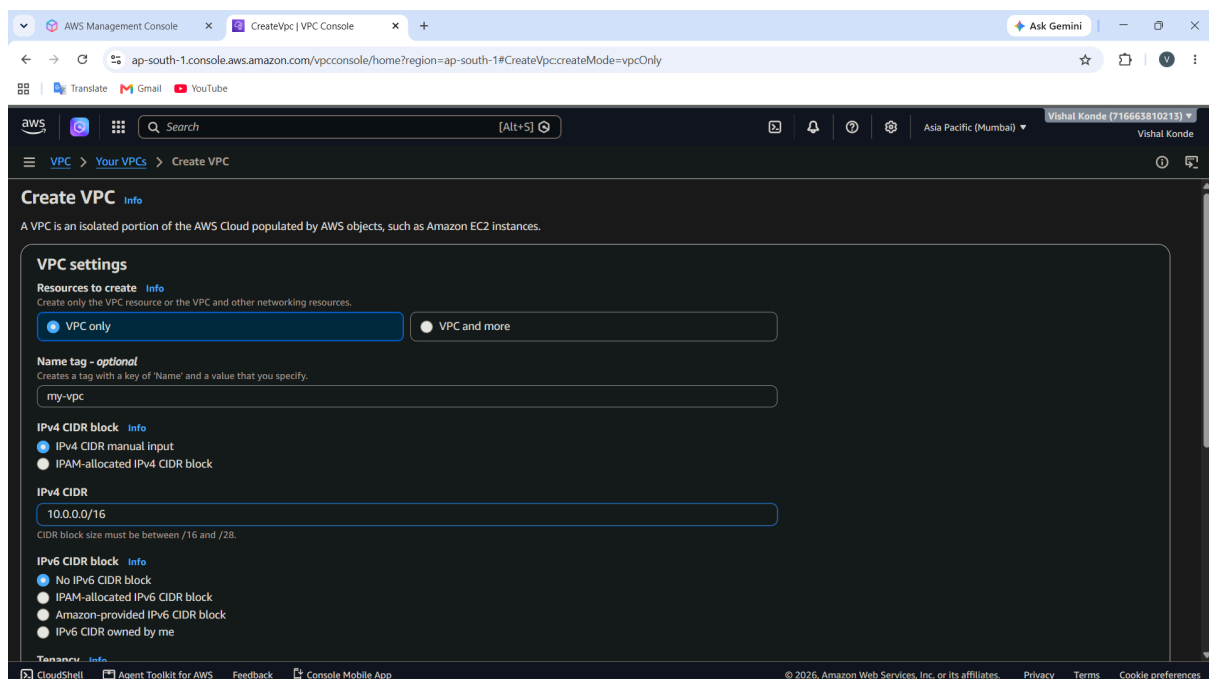
1 . Login to AWS Console



2 . Create VPC

Configuration :

Cidr - 10.0.0.0/16



3 . Create Subnets

1 . Public Subnet :

Cidr : 10.0.0.0/24

Region : ap-south-1a

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
my-public-subnet
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.0.0/24 256 IPs

Tags - optional
Key Value - optional
Name my-public-subnet Remove

2 . Private Subnet-01 :

Cidr : 10.0.1.0/24

Region : ap-south-1b

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
my-private-01
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

Tags - optional
Key Value - optional
Name my-private-01 Remove

Add new tag
You can add 49 more tags.

Remove

3 . Private Subnet-02 :

Cidr : 10.0.2.0/24

Region : ap-south-1c

Subnet 3 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
my-private-02
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-s1-az2 (ap-south-1c)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

Tags - optional
Key Value - optional
Name my-private-02
Add new tag
Remove

Subnets Created Successfully

You have successfully created 3 subnets: subnet-05edf6a148c8ce1d8, subnet-085f7179714de00bd, subnet-0249769761630b432

Subnets (3) Info
Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
my-private-01	subnet-085f7179714de00bd	Available	vpc-079b90e5d5c306a9a my-...	Off	10.0.1.0/24
my-public-subnet	subnet-05edf6a148c8ce1d8	Available	vpc-079b90e5d5c306a9a my-...	Off	10.0.0.0/24
my-private-02	subnet-0249769761630b432	Available	vpc-079b90e5d5c306a9a my-...	Off	10.0.2.0/24

4 . Create Internet Gateway

The screenshot shows the AWS Management Console interface for creating a new internet gateway. The browser address bar indicates the URL: `ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateInternetGateway`. The console header shows the user is logged in as Vishal Konde in the Asia Pacific (Mumbai) region. The breadcrumb navigation is `VPC > Internet gateways > Create internet gateway`. The main heading is **Create internet gateway** with an **Info** link. A descriptive text states: "An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below." The form is divided into two sections: **Internet gateway settings** and **Tags - optional**. In the first section, the **Name tag** field is populated with `my-igw`. The second section explains that tags are labels for AWS resources and shows a table with one tag: `my-igw`. At the bottom right, there are **Cancel** and **Create internet gateway** buttons.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

`my-igw`

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Key	Value - optional	
<code>my-igw</code>	<code>my-igw</code>	Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

Attach VPC

The screenshot shows the AWS Management Console interface for attaching an existing internet gateway to a VPC. The browser address bar indicates the URL: `ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#AttachInternetGateway:internetGatewayId=igw-085e7a7c98b1f35da`. The console header shows the user is logged in as Vishal Konde in the Asia Pacific (Mumbai) region. The breadcrumb navigation is `VPC > Internet gateways > Attach to VPC (igw-085e7a7c98b1f35da)`. A green notification banner at the top states: "The following internet gateway was created: igw-085e7a7c98b1f35da - my-igw. You can now attach to a VPC to enable the VPC to communicate with the internet." The main heading is **Attach to VPC (igw-085e7a7c98b1f35da)** with an **Info** link. The **VPC** section explains: "Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below." The **Available VPCs** section shows a search bar with the text `vpc-079b90e5d5c306a9a`. Below this, there is a section for the **AWS Command Line Interface command**. At the bottom right, there are **Cancel** and **Attach internet gateway** buttons.

[Attach to a VPC](#) [X](#)

Attach to VPC (igw-085e7a7c98b1f35da) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

`vpc-079b90e5d5c306a9a`

AWS Command Line Interface command

[Cancel](#) [Attach internet gateway](#)

5 . Create Public Route table for Internet Gateway and Public Subnet.

The screenshot shows the 'Create route table' page in the AWS Management Console. The page is titled 'Create route table' and includes a breadcrumb trail: VPC > Route tables > Create route table. The main content area is divided into two sections: 'Route table settings' and 'Tags'.

Route table settings

- Name - optional:** A text input field containing 'my-rt-igw'.
- VPC:** A dropdown menu showing 'vpc-079b90e5d5c306a9a (my-vpc)'.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X Q my-rt-igw X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Route Internet Gateway

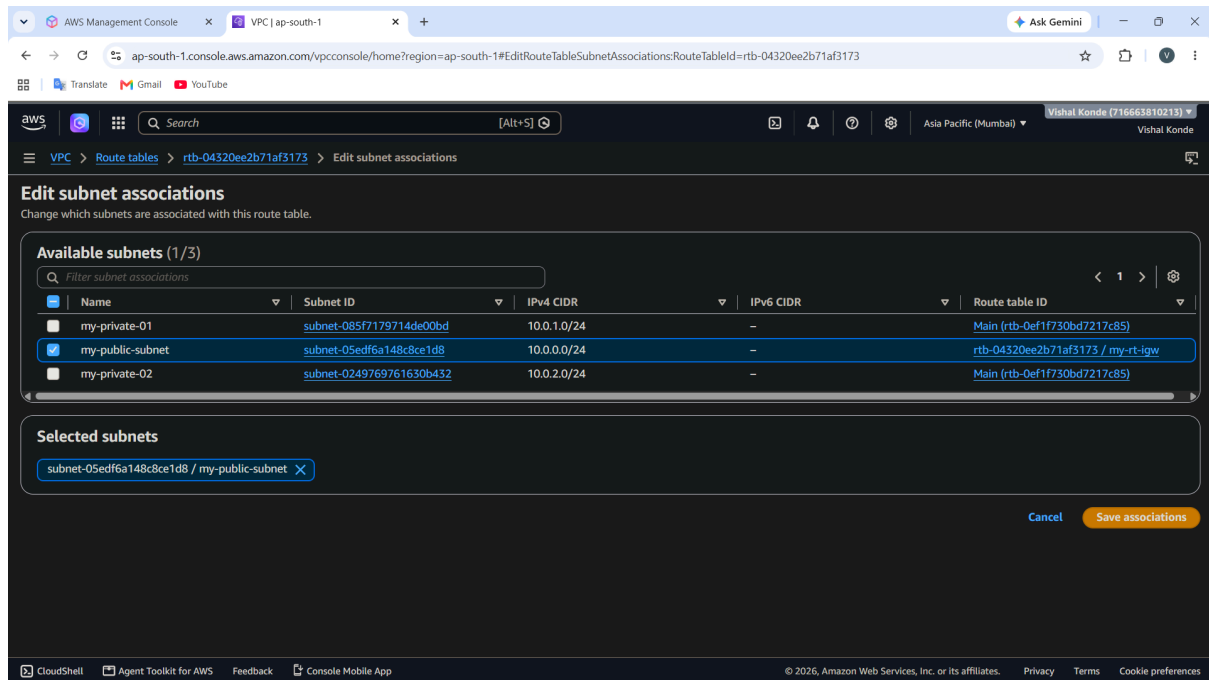
The screenshot shows the 'Edit routes' page in the AWS Management Console. The page is titled 'Edit routes: rtb-04320ee2b71af3173 / my-rt-igw'. The main content area is a table with columns: Destination, Target, Status, Propagated, and Route Origin.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

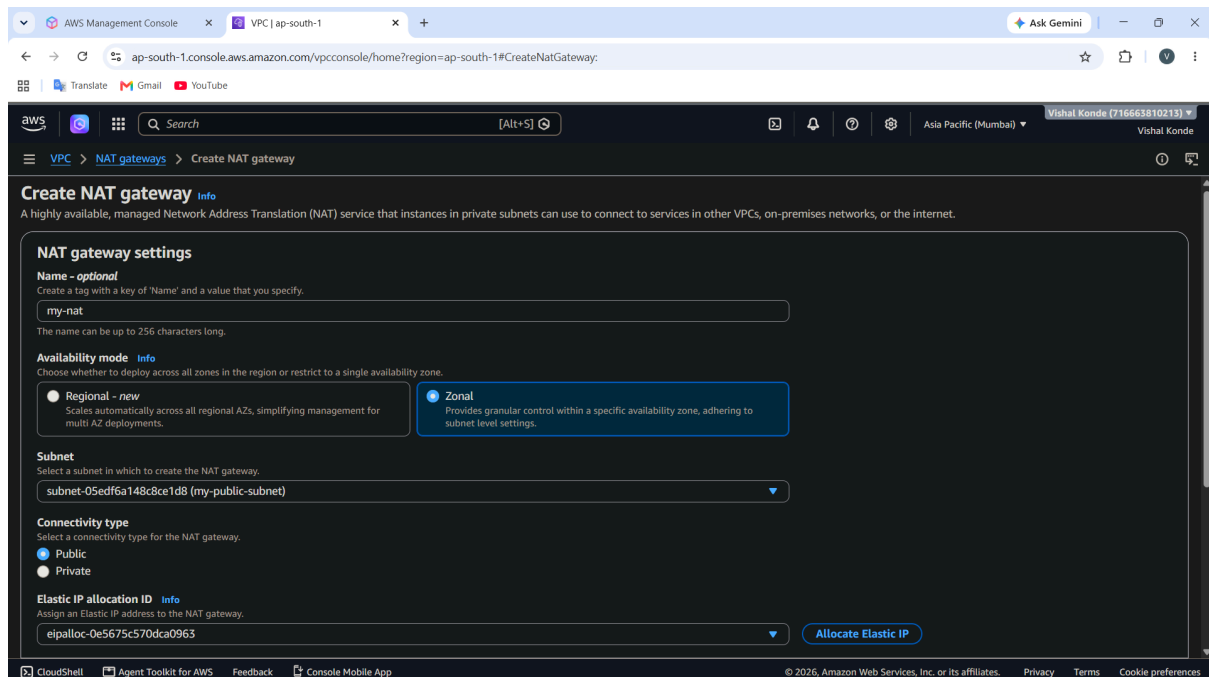
Add route

Cancel Preview Save changes

Associate Public Subnet to Route Table

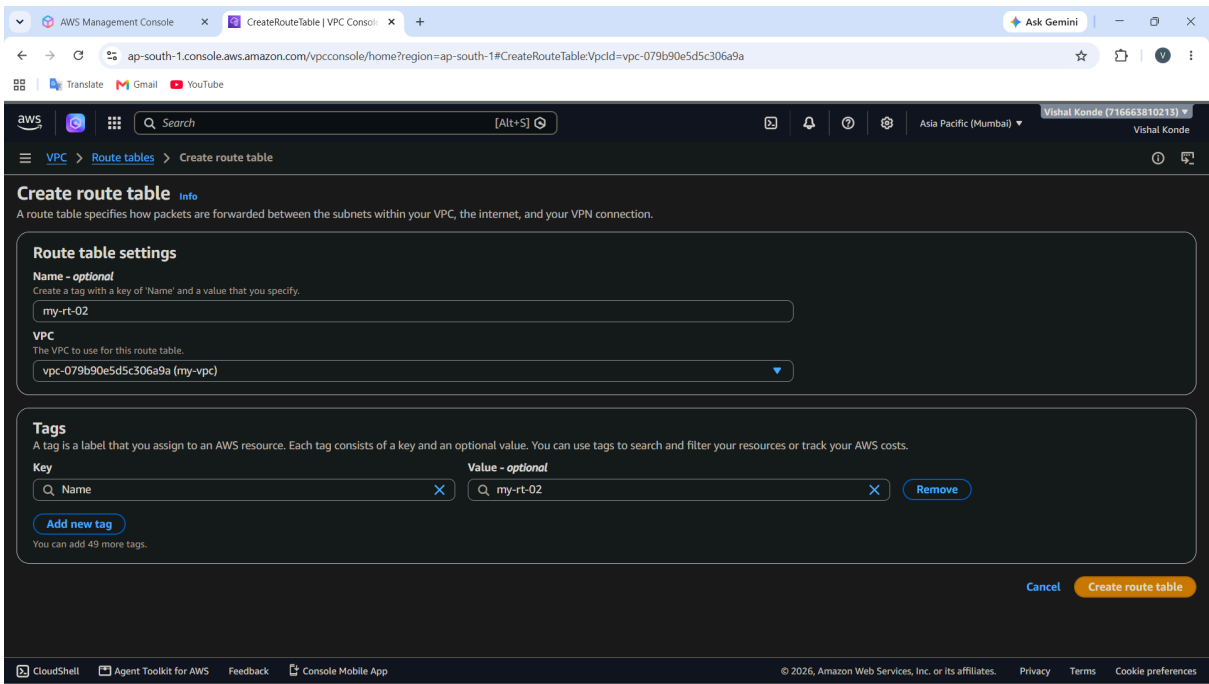


6 . Create Nat Gateway

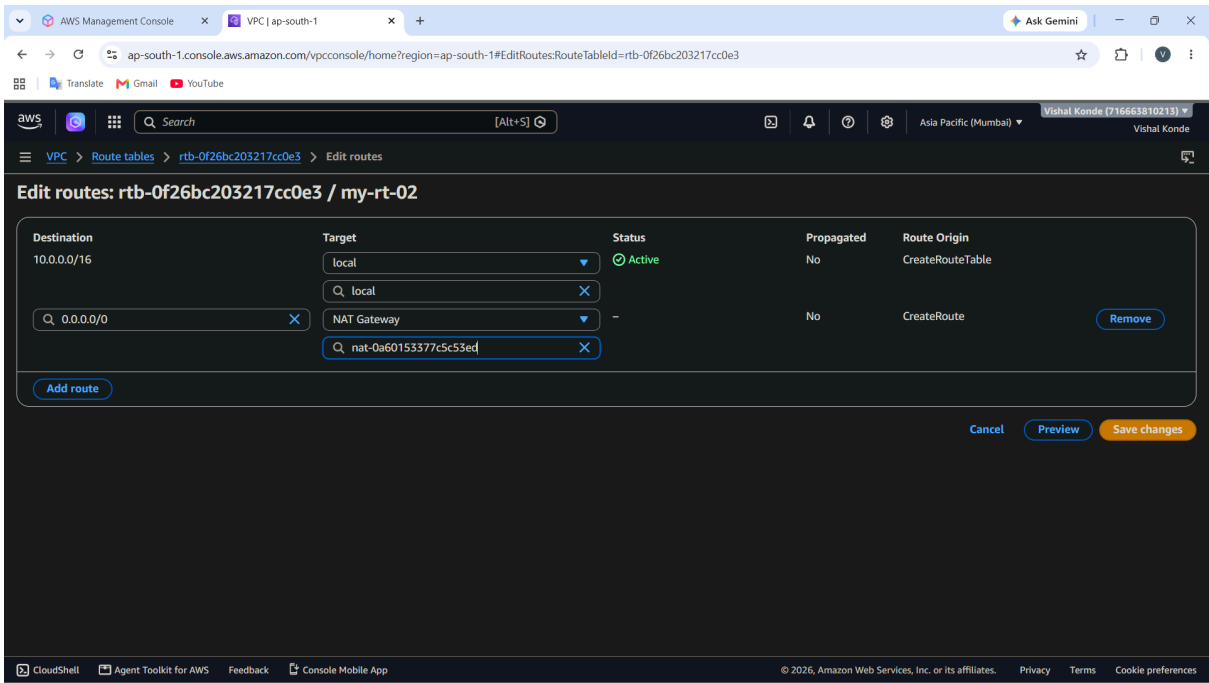


Associate Elastic IP

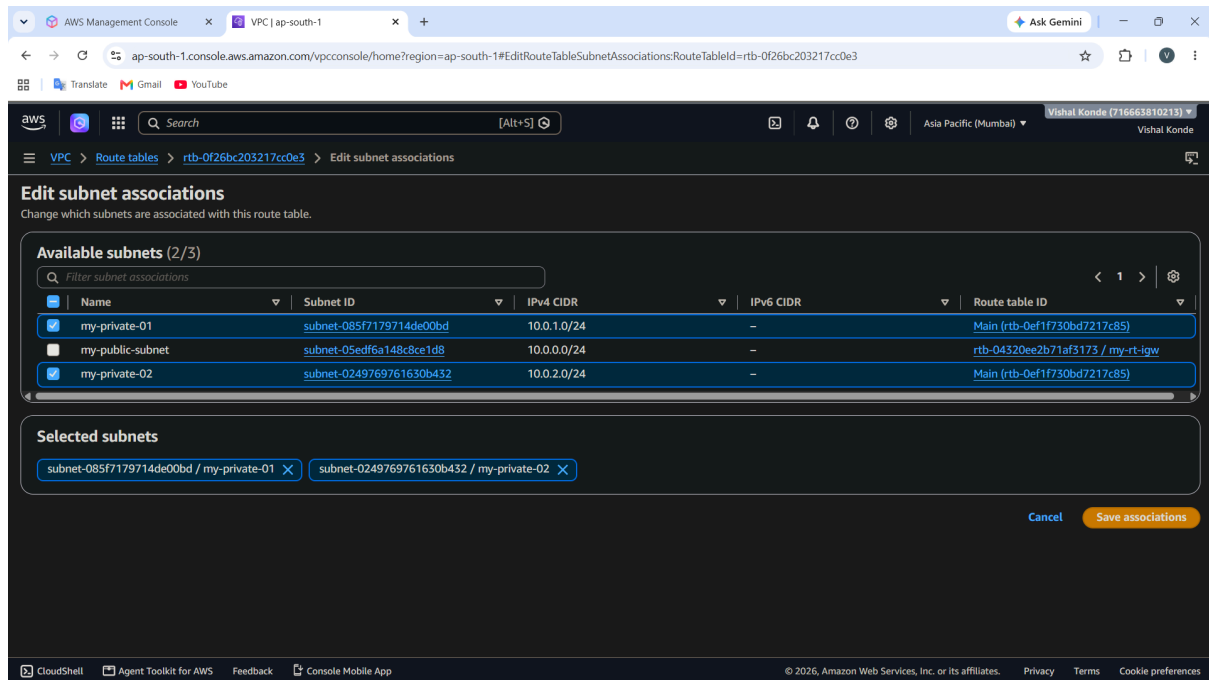
7 . Create Route Table for nat Gateway



Route Nat Gateway



Associate Private Subnets

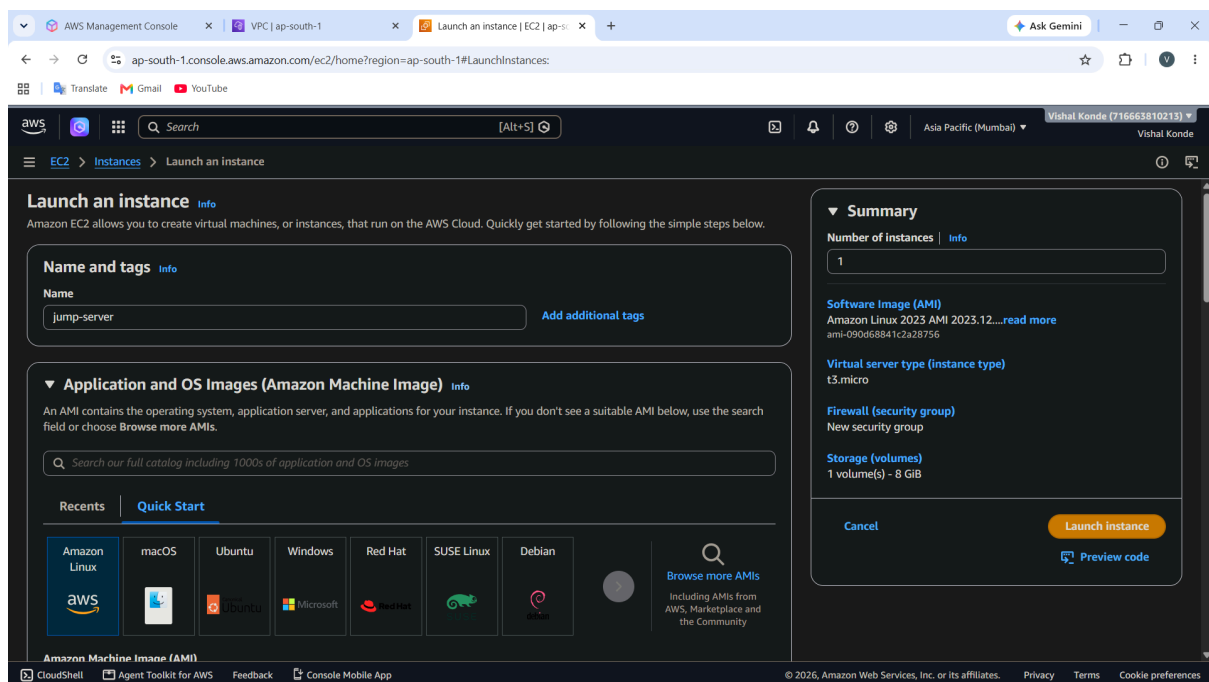


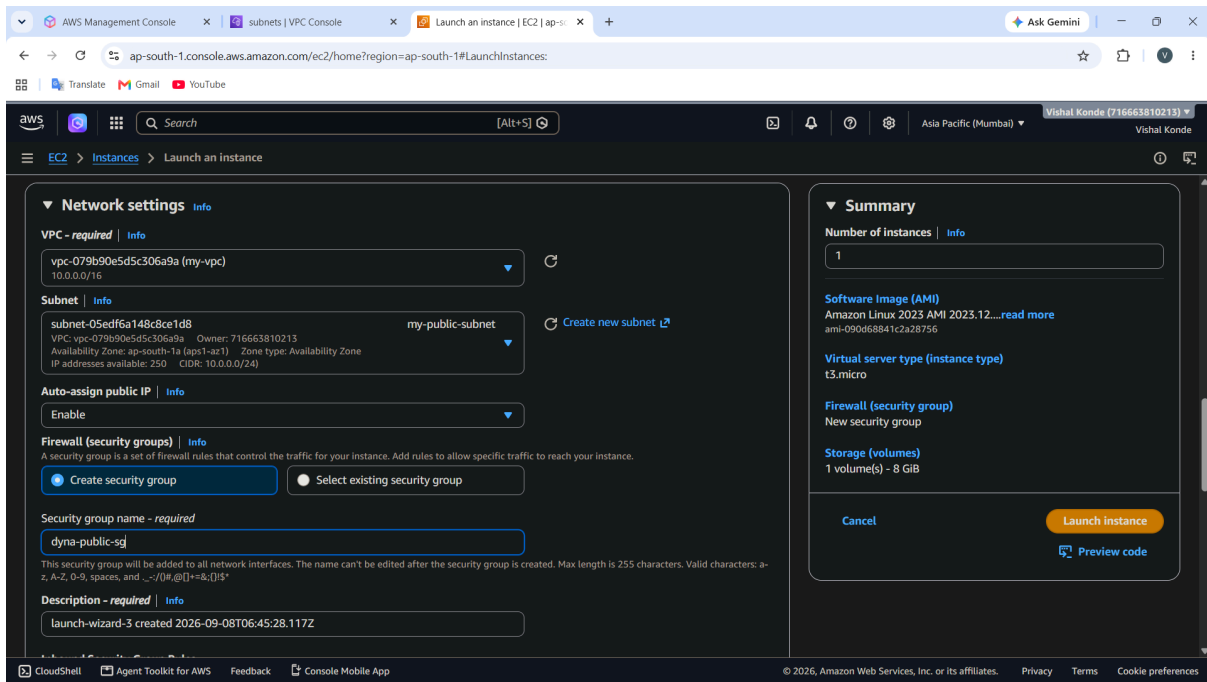
8 . Create EC2 Instance in Public Subnet

Config : Name - Jump-server

Subnet - Public-Subnet

SG - dyna-public-sg with http(80) allowed



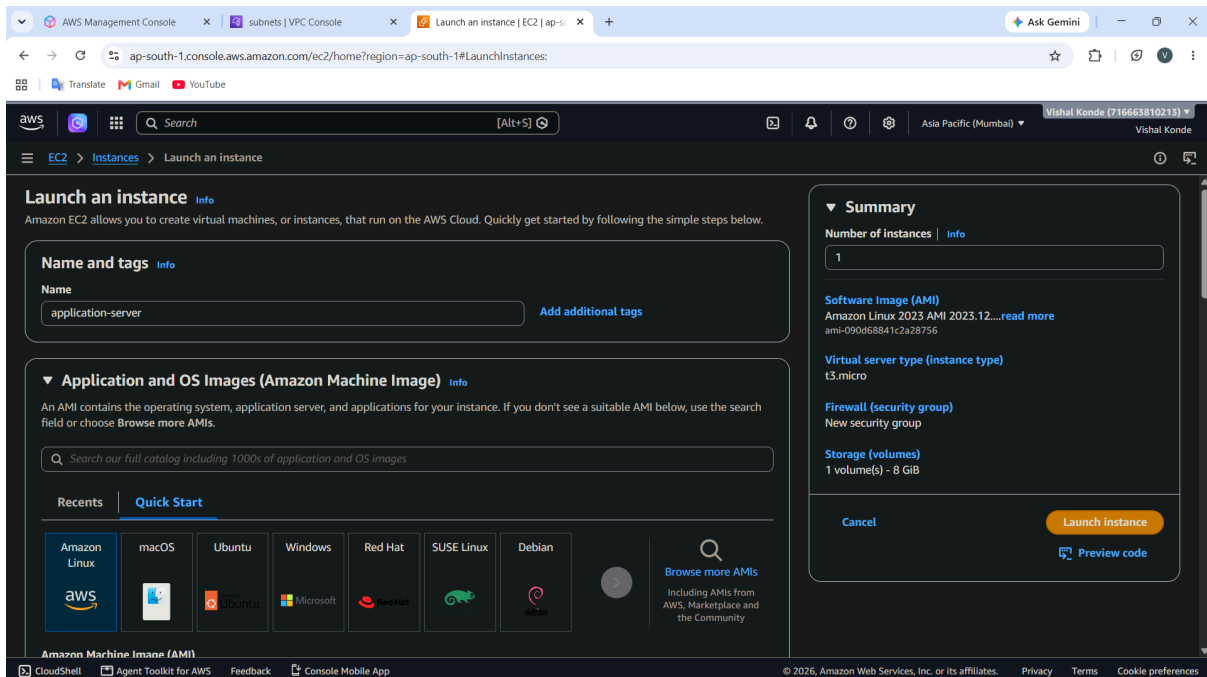


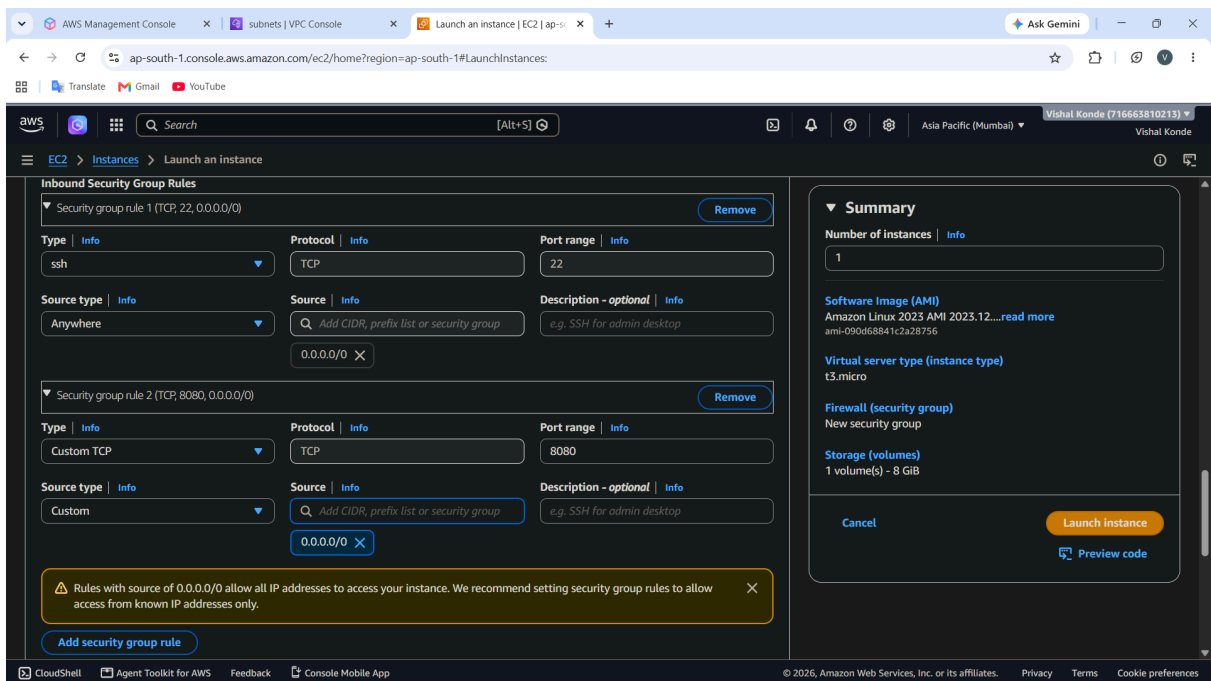
9 . Create EC2 Instance in private-subnet -01

Config : Name - Application-server

Subnet - Private-Subnet-01

SG - dyna-private-01 with 8080 allowed



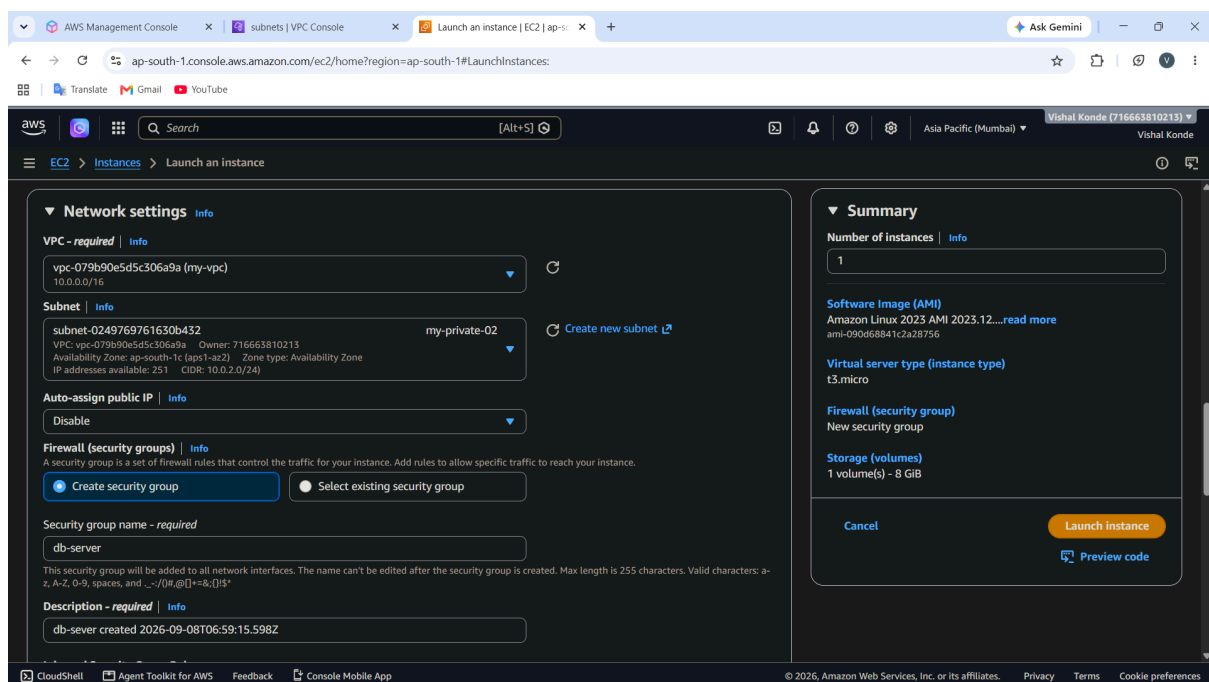
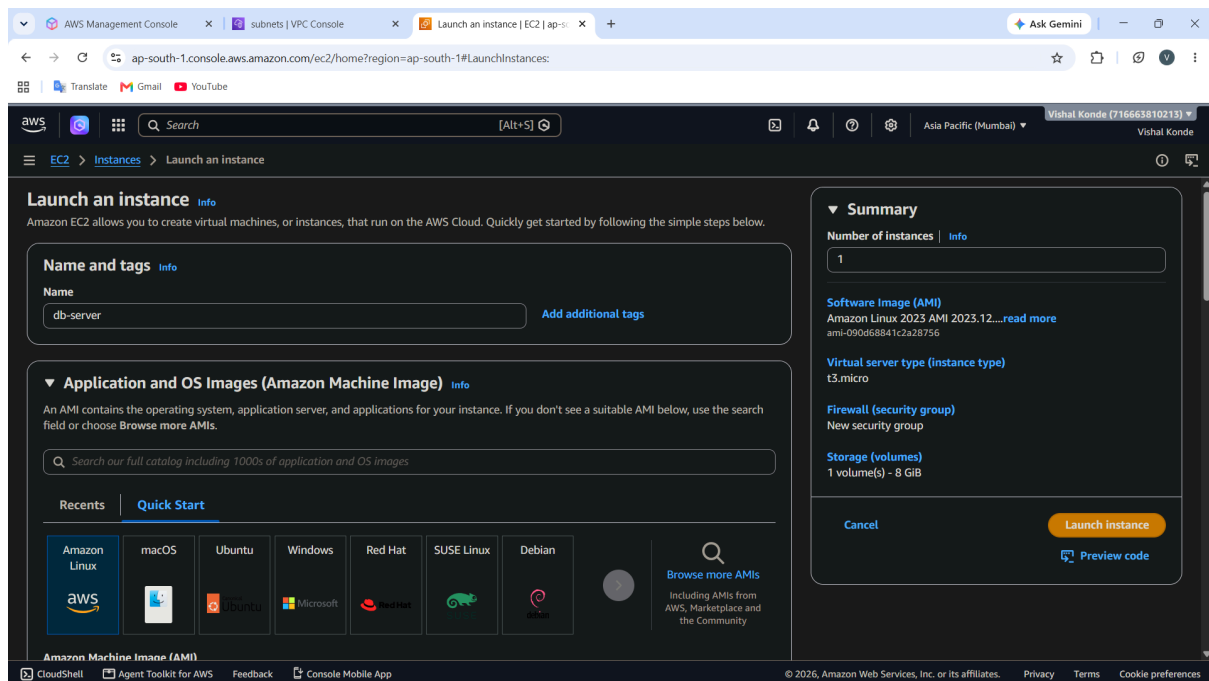


10 . Create EC3 Instance in private-subnet-02

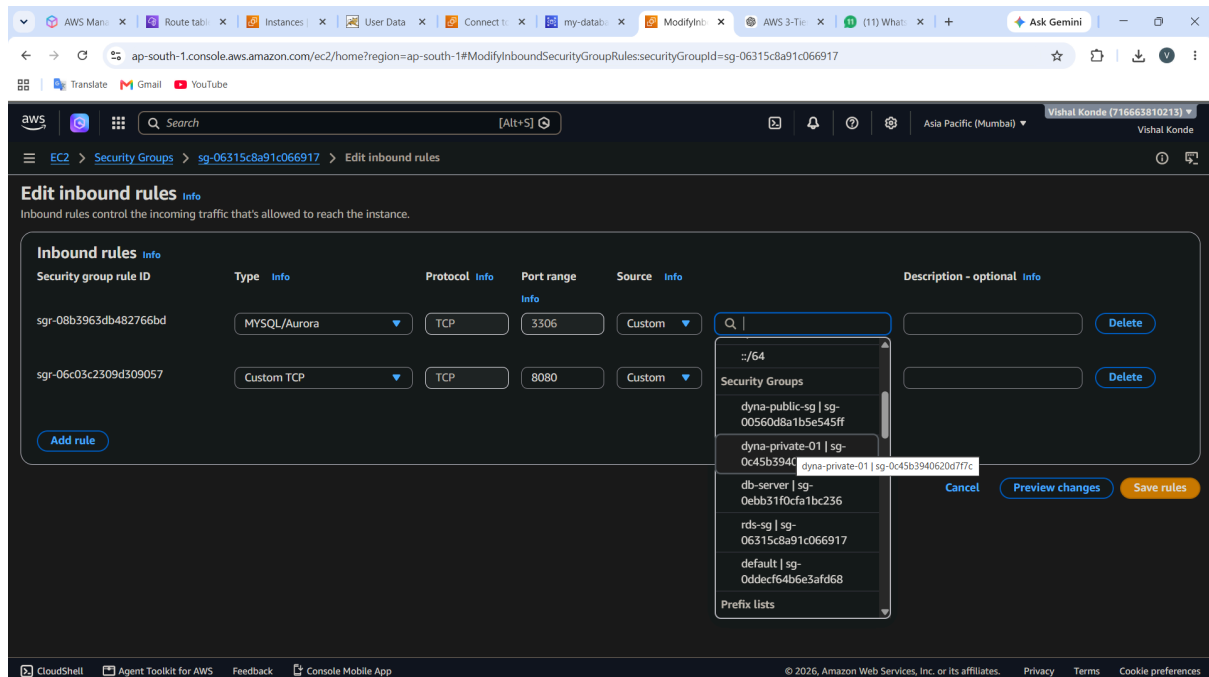
Config : Name - db-server

Subnet - Private-Subnet-02

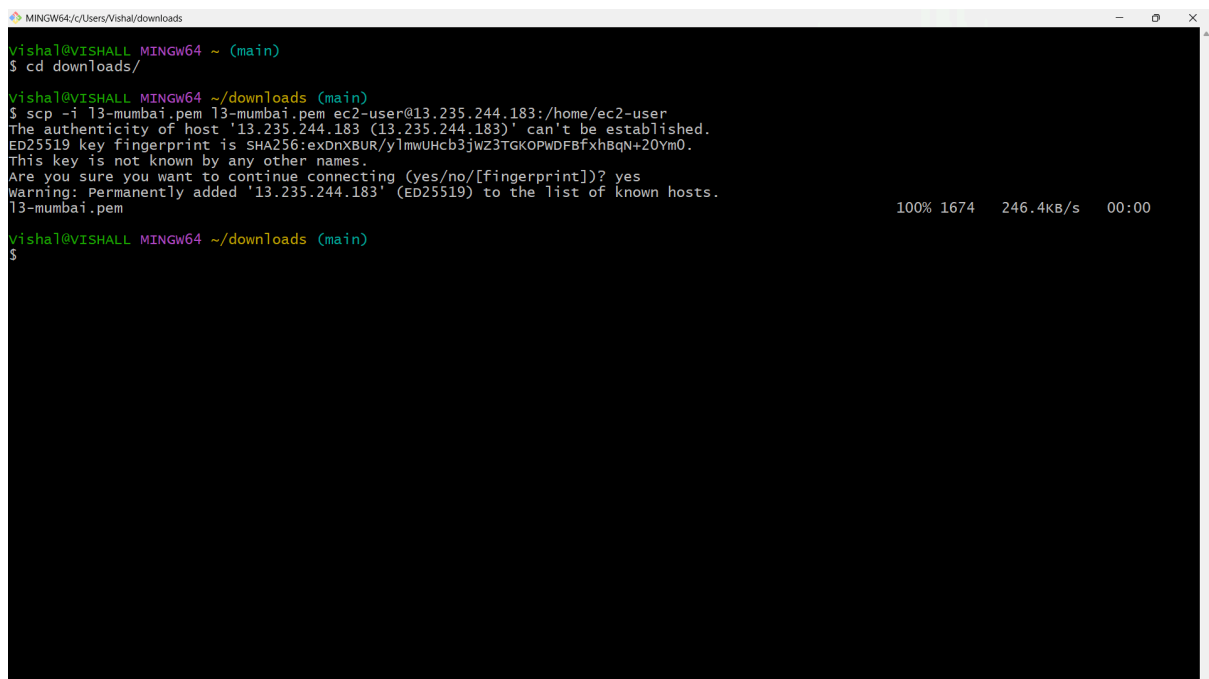
SG - db-server with 3306 allowed



3306 Should be Allowed with Source coming from application-server's Security group



11. Copy Key-pair To Remote Jump Server from Local Machine's downloads



12 . Connect Public Subnet

[illegible]

13 . Install Nginx Server in Jump-server instance .

```

root@ip-10-0-0-200-:~# yum install nginx -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

=====
Package                                Architecture      Version            Repository          Size
=====
Installing:
nginx                                  x86_64            1:1.30.4-1.amzn2023.0.1  amazonlinux        34 k
Installing dependencies:
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3  amazonlinux        19 k
gperftools-libs                      x86_64            2.9.1-1.amzn2023.0.3    amazonlinux        308 k
libunwind                            x86_64            1.4.0-5.amzn2023.0.3    amazonlinux        66 k
nginx-core                           x86_64            1:1.30.4-1.amzn2023.0.1  amazonlinux        711 k
nginxfilesystem                      noarch            1:1.30.4-1.amzn2023.0.1  amazonlinux        10 k
nginx-mimetypes                      noarch            2.1.49-3.amzn2023.0.3    amazonlinux        21 k
=====
Transaction Summary
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                1.6 MB/s | 66 kB | 00:00
(2/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm    428 kB/s | 19 kB | 00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm         5.8 MB/s | 308 kB | 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm                  1.2 MB/s | 34 kB | 00:00
(5/7): nginxfilesystem-1.30.4-1.amzn2023.0.1.noarch.rpm        400 kB/s | 10 kB | 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm             16 MB/s | 711 kB | 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm        845 kB/s | 21 kB | 00:00
=====
Total                                                                    8.3 MB/s | 1.1 MB | 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Running scriptlet: nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch
  Installing : nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch
  Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

```

Start the nginx .

```
root@ip-10-0-0-200:~#
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 2/7
Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
Verifying : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 4/7
Verifying : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/7
Verifying : nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch 6/7
Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 7/7

Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch
gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
nginx-1:1.30.4-1.amzn2023.0.1.x86_64
nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch

complete!
[root@ip-10-0-0-200 ~]# systemctl start nginx.service
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]#
[root@ip-10-0-0-200 ~]# systemctl status nginx.service
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 07:09:16 UTC; 15s ago
     Process: 26428 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 26429 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 26430 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 26431 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.5M
        CPU: 58ms
    CGroup: /system.slice/nginx.service
            └─26431 "nginx: master process /usr/sbin/nginx"
              └─26432 "nginx: worker process"
                └─26433 "nginx: worker process"

Sep 08 07:09:16 ip-10-0-0-200.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server:
Sep 08 07:09:16 ip-10-0-0-200.ap-south-1.compute.internal nginx[26429]: nginx: the configuration file /etc/nginx/nginx.conf syntax is correct
Sep 08 07:09:16 ip-10-0-0-200.ap-south-1.compute.internal nginx[26429]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 08 07:09:16 ip-10-0-0-200.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
lines 1-19/19 (END)
```

14 . SSH/Connect to Application Server .

Download Tomcat Apache and Unzip the Folder in opt .

```
root@ip-10-0-1-204:~#
curl: (2) no URL specified
curl: try 'curl --help' for more information
[root@ip-10-0-1-204 ~]# curl -o https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
% Total % Received % Xferd Average Speed Time Time Time
Dload Upload Total Spent Left Speed
100 12869k 100 12869k 0 0 205.2M 0 --:--:-- --:--:-- --:--:-- 209.4M
[root@ip-10-0-1-204 ~]# ls
apache-tomcat-9.0.121.tar.gz
[root@ip-10-0-1-204 ~]# tar -xzf apache-tomcat-9.0.121.tar.gz -C/opt
apache-tomcat-9.0.121/conf/
apache-tomcat-9.0.121/conf/catalina.policy
apache-tomcat-9.0.121/conf/catalina.properties
apache-tomcat-9.0.121/conf/context.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xml
apache-tomcat-9.0.121/conf/jaspic-providers.xsd
apache-tomcat-9.0.121/conf/logging.properties
apache-tomcat-9.0.121/conf/server.xml
apache-tomcat-9.0.121/conf/tomcat-users.xml
apache-tomcat-9.0.121/conf/tomcat-users.xsd
apache-tomcat-9.0.121/conf/web.xml
apache-tomcat-9.0.121/bin/
apache-tomcat-9.0.121/lib/
apache-tomcat-9.0.121/logs/
apache-tomcat-9.0.121/temp/
apache-tomcat-9.0.121/webapps/
apache-tomcat-9.0.121/webapps/ROOT/
apache-tomcat-9.0.121/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/
apache-tomcat-9.0.121/webapps/docs/META-INF/
apache-tomcat-9.0.121/webapps/docs/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.121/webapps/docs/annotationapi/
apache-tomcat-9.0.121/webapps/docs/api/
apache-tomcat-9.0.121/webapps/docs/appdev/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.121/webapps/docs/appdev/sample/web/images/
```

15 . Download the student registration Application in /opt/apache/webapps folder

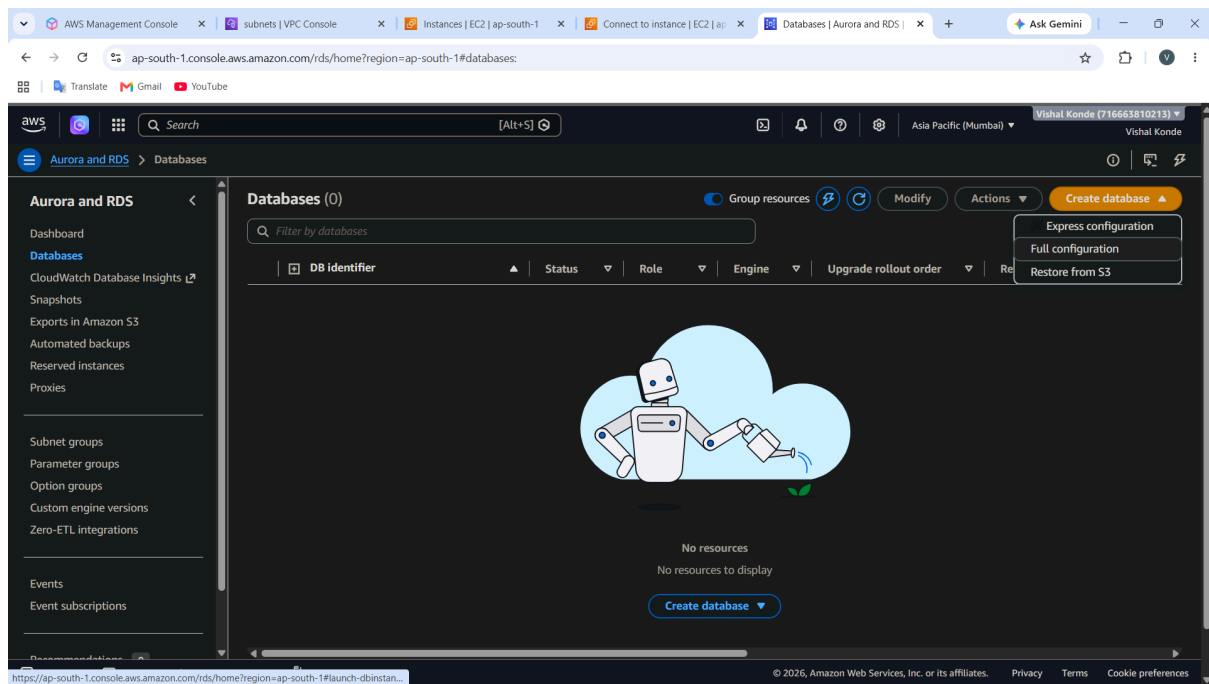
```
root@ip-10-0-1-204:/
apache-tomcat-9.0.121/webapps/manager/images/favicon.ico
apache-tomcat-9.0.121/webapps/manager/images/tomcat.svg
apache-tomcat-9.0.121/webapps/manager/index.jsp
apache-tomcat-9.0.121/webapps/manager/status.xsd
apache-tomcat-9.0.121/webapps/manager/xform.xsl
apache-tomcat-9.0.121/bin/catalina.sh
apache-tomcat-9.0.121/bin/ciphers.sh
apache-tomcat-9.0.121/bin/configtest.sh
apache-tomcat-9.0.121/bin/daemon.sh
apache-tomcat-9.0.121/bin/digest.sh
apache-tomcat-9.0.121/bin/makebase.sh
apache-tomcat-9.0.121/bin/setclasspath.sh
apache-tomcat-9.0.121/bin/shutdown.sh
apache-tomcat-9.0.121/bin/startup.sh
apache-tomcat-9.0.121/bin/tool-wrapper.sh
apache-tomcat-9.0.121/bin/version.sh
[root@ip-10-0-1-204 ~]# ls
apache-tomcat-9.0.121.tar.gz
[root@ip-10-0-1-204 ~]# cd /opt
[root@ip-10-0-1-204 opt]# ls
apache-tomcat-9.0.121
[root@ip-10-0-1-204 opt]# cd apache-tomcat-9.0.121/
[root@ip-10-0-1-204 apache-tomcat-9.0.121]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-10-0-1-204 apache-tomcat-9.0.121]# cd webapps
[root@ip-10-0-1-204 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-10-0-1-204 webapps]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war
% Total % Received % Xferd Average Speed Time Time Current
100 89423 100 89423 0 0 71873 0 0:00:01 0:00:01 --:--:-- 71883
[root@ip-10-0-1-204 webapps]# ls
ROOT docs examples host-manager manager student.war
[root@ip-10-0-1-204 webapps]# cd ..
[root@ip-10-0-1-204 apache-tomcat-9.0.121]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-10-0-1-204 apache-tomcat-9.0.121]# cd ..
[root@ip-10-0-1-204 opt]# cd ..
[root@ip-10-0-1-204 /]# yum install java -y
```

16 . Install Java And Start the Tomcat using [./catalina.sh](#) start

```
ec2-user@ip-10-0-0-200:~
Verifying : pixman-0.43.4-1.amzn2023.0.4.x86_64
Verifying : xml-common-0.6.3-56.amzn2023.0.2.noarch
34/35
Installed:
alsa-lib-1.2.7.2-1.amzn2023.0.4.x86_64
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
fontconfig-devel-2.13.94-2.amzn2023.0.2.x86_64
fontconfig-filesystem-1:2.0.5-12.amzn2023.0.2.noarch
giflib-5.2.1-9.amzn2023.0.4.x86_64
google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
java-26-amazon-corretto-headless-1:26.0.2+11-1.amzn2023.1.x86_64
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch
cairo-1.18.0-4.amzn2023.0.3.x86_64
dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
freetype-2.13.2-5.amzn2023.0.2.x86_64
google-noto-fonts-common-20240401-1.amzn2023.0.2.noarch
graphite2-1.3.14-7.amzn2023.0.3.x86_64
java-26-amazon-corretto-1:26.0.2+11-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libxt-1.3.0-3.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libpng-2:1.6.37-10.amzn2023.0.13.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64

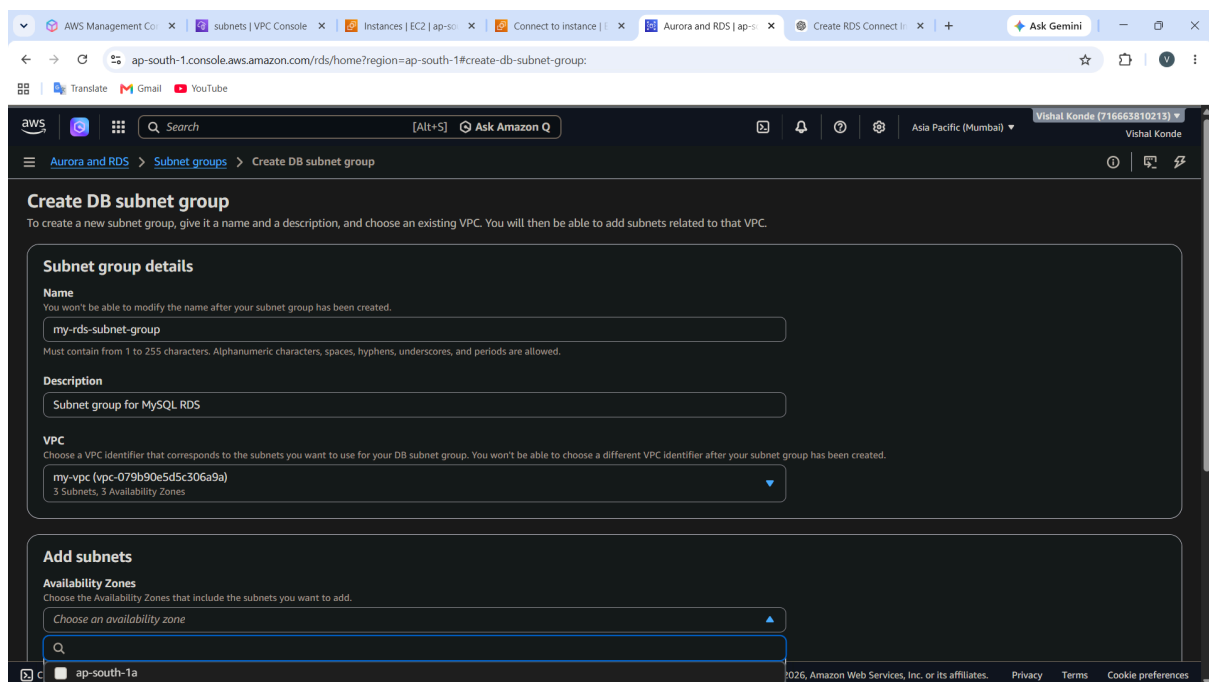
complete!
[root@ip-10-0-1-204 /]# cd opt
[root@ip-10-0-1-204 opt]# cd apache-tomcat-9.0.121/
[root@ip-10-0-1-204 apache-tomcat-9.0.121]# cd bin
[root@ip-10-0-1-204 apache-tomcat-9.0.121/bin]# ls
bootstrap.jar ciphers.sh commons-daemon-native.tar.gz daemon.sh digest.sh setclasspath.bat startup.sh version.sh
catalina-tasks.xml commons-daemon.jar digest.sh shutdown.bat tomcat-juli.jar version.sh
catalina.bat commons-daemon-native.tar.gz digest.sh shutdown.bat tomcat-native.tar.gz version.sh
catalina.sh configtest.sh makebase.sh startup.bat tool-wrapper.bat tool-wrapper.sh
ciphers.bat configtest.sh makebase.sh startup.bat tool-wrapper.bat tool-wrapper.sh
[root@ip-10-0-1-204 apache-tomcat-9.0.121/bin]# ./catalina.sh start
Using CATALINA_BASE: /opt/apache-tomcat-9.0.121
Using CATALINA_HOME: /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

17 . Navigate to RDS Dashboard

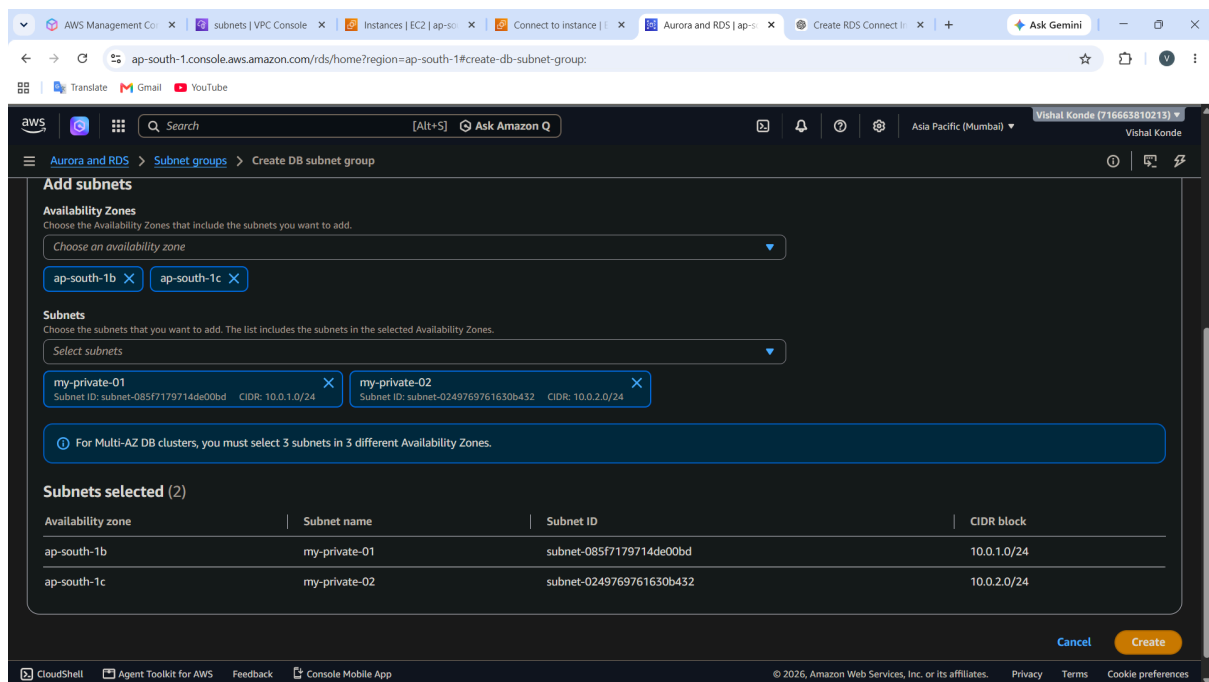


18 . Create DB Subnet Group .

Name : my-rds-subnet-group

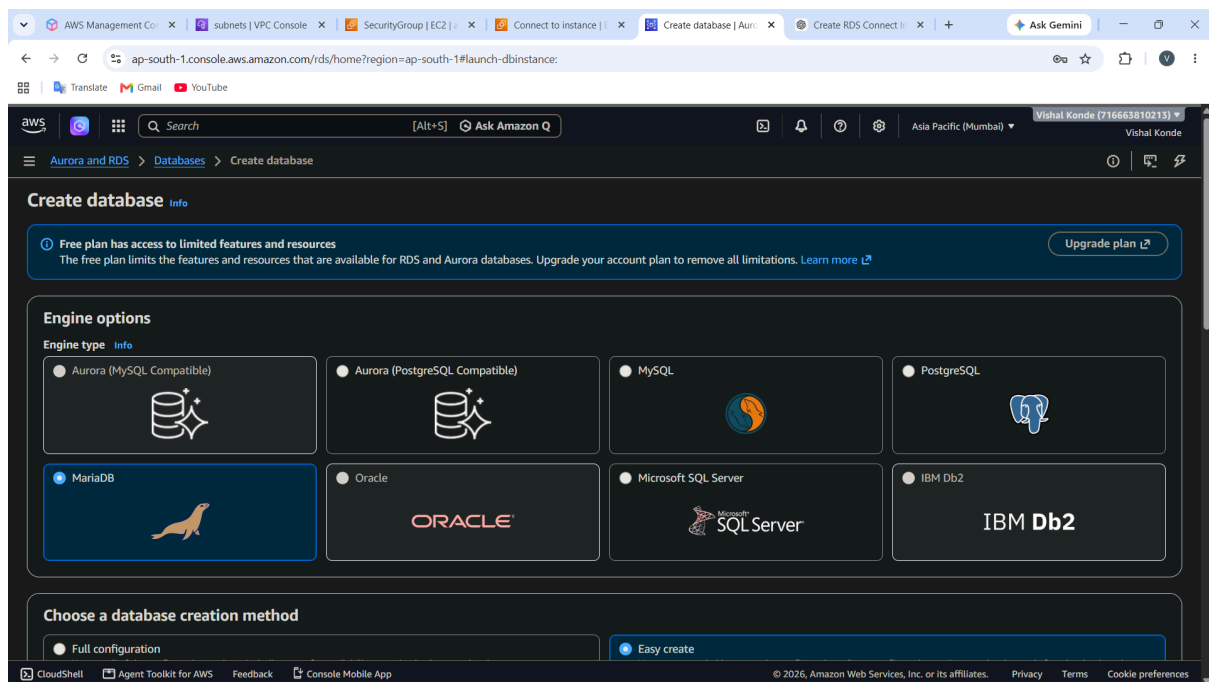


Select Availability Zones and both Private Subnets:



19 . Create Database

Engine : MariaDB



Specify Name , Username , Password

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

Engine version
MariaDB 11.8.8

DB instance identifier **Info**
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
my-database
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username **Info**
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
☒ Self managed
Create your own password or have RDS create a password that you manage.

Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password **Info**
Password strength **Very weak**
Minimum constraint: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ! * @

Select VPC , DB-Subnet-Group

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
my-vpc (vpc-079b90e5d5c306a9a)
3 Subnets, 3 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

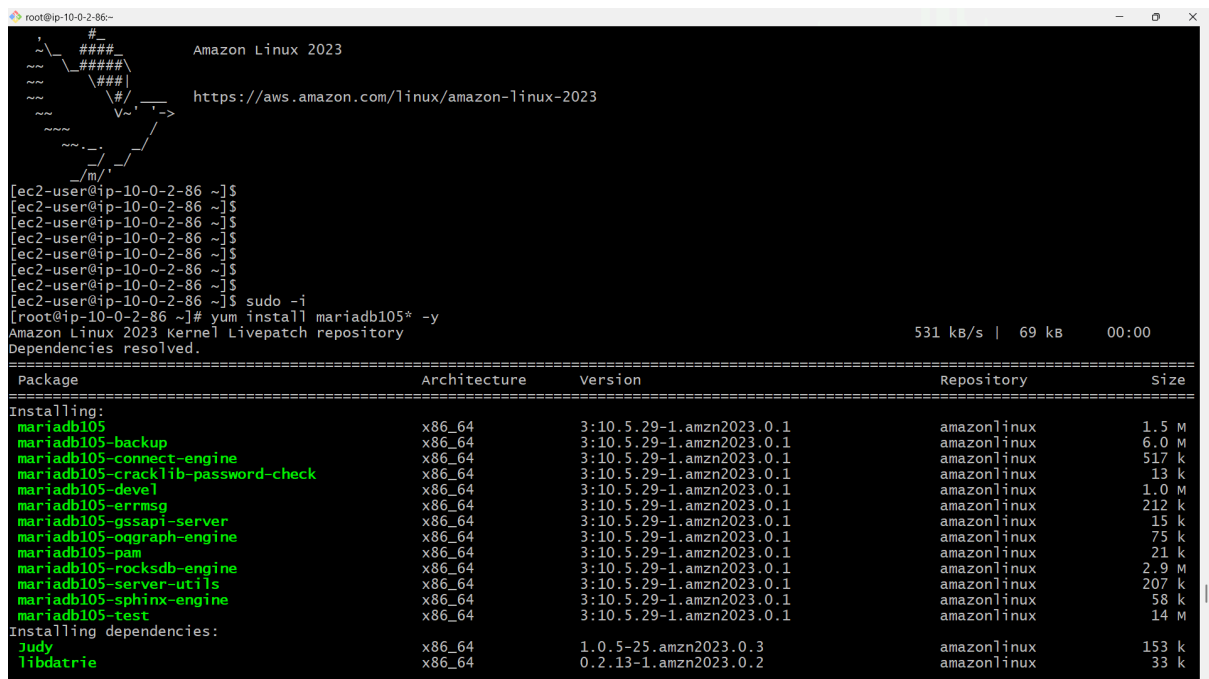
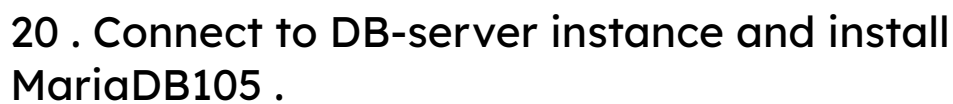
DB subnet group **Info**
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
my-rds-subnet-group
2 Subnets, 2 Availability Zones

Public access **Info**
☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) **Info**
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
☒ Choose existing
Choose existing VPC security groups
☐ Create new
Create new VPC security group

Existing VPC security groups
Choose one or more options

It Gives 2 Connection Links which we have copy paste after connecting and installing MariaDB in instance .



Start the MariaDB service

```

root@ip-10-0-2-86:~#
perl -Term-Table-0.015-6.amzn2023.0.2.noarch
perl -Tie-4.6-477.amzn2023.0.9.noarch
perl -Unicode-LineBreak-2019.001-9.amzn2023.0.4.x86_64
perl -lib-0.65-477.amzn2023.0.9.x86_64
perl-threads-shared-1.61-458.amzn2023.0.3.x86_64
sphinx-2.2.11-24.amzn2023.0.4.x86_64
perl-Test-Simple-3.1.302219-2.amzn2023.0.1.noarch
perl -Time-1.03-477.amzn2023.0.9.noarch
perl-deprecate-0.04-477.amzn2023.0.9.noarch
perl-threads-1.2.25-458.amzn2023.0.4.x86_64
sombok-2.4.0-14.amzn2023.0.2.x86_64
zlib-devel-1.2.11-33.amzn2023.0.6.x86_64

Complete!
[root@ip-10-0-2-86 ~]# systemctl start mariadb.service
[root@ip-10-0-2-86 ~]# systemctl status mariadb.service
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 12:01:11 UTC; 7s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 26520 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 26579 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 28819 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
  Main PID: 28616 (mariadb)
    Status: "Taking your SQL requests now..."
     Tasks: 21 (limit: 1059)
    Memory: 96.3M
       CPU: 583ms
    CGroup: /system.slice/mariadb.service
            └─28616 /usr/libexec/mariadb --basedir=/usr

Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: The second is mysql@localhost, it has no passw
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: you need to be the system 'mysql' user to conn
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: After connecting you can set the password, if
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: able to connect as any of these users with a p
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: see the MariaDB Knowledgebase at https://maria
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: Please report any problems at https://mariadb
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: The latest information about MariaDB is availa
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: Consider joining MariaDB's strong and vibrant
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: https://mariadb.org/get-involved/
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[root@ip-10-0-2-86 ~]# systemctl enable mariadb.service
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-10-0-2-86 ~]#

```

21 . Copy and Paste the Connection Links and Enter Password .

Database will be Connected Successfully.

```
root@ip-10-0-2-86:~#  
Process: 26520 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)  
Process: 26579 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)  
Process: 28819 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)  
Main PID: 28616 (mariadb)  
Status: "Taking your SQL requests now..."  
Tasks: 21 (limit: 1059)  
Memory: 96.3M  
CPU: 583ms  
CGroup: /system.slice/mariadb.service  
└─28616 /usr/libexec/mariadb --basedir=/usr  
  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: The second is mysql@localhost, it has no passw  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: you need to be the system 'mysql' user to conn  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: After connecting you can set the password, if  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: able to connect as any of these users with a p  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: please see the MariaDB Knowledgebase at https://maria  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: Please report any problems at https://mariadb.o  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: The latest information about MariaDB is availab  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: Consider joining MariaDB's strong and vibrant  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: https://mariadb.org/get-involved/  
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.  
[root@ip-10-0-2-86 ~]# systemctl enable mariadb.service  
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.  
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.  
[root@ip-10-0-2-86 ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem  
% Total % Received % Xferd Average Speed Time Time Current  
100 165408 100 165408 0 0 230073 0 ----- :----- 230052  
[root@ip-10-0-2-86 ~]# mysql -h my-database.cvqcikarn6.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ss  
l-ca=global-bundle.pem  
Enter password:  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 47  
Server version: 11.8.8-MariaDB managed by https://aws.amazon.com/rds/  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> |
```

22 . Create Database and Table .

```

root@ip-10-0-2-86:~#
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal mariadb-prepare-db-dir[27587]: https://mariadb.org/get-involved/
Sep 08 12:01:11 ip-10-0-2-86.ap-south-1.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[root@ip-10-0-2-86 ~]# systemctl enable mariadb.service
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-10-0-2-86 ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 165408 100 165408    0     0 230073      0 --:--:-- --:--:-- --:--:-- 230052
[root@ip-10-0-2-86 ~]# mysql -h my-database.cvqyci0karn6.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 47
Server version: 11.8.8-MariaDB managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database studentapp;
Query OK, 1 row affected (0.006 sec)

MariaDB [(none)]> show databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'databases' at line 1
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| performance_schema |
| studentapp |
| sys |
+-----+
6 rows in set (0.002 sec)

MariaDB [(none)]> |

```

23 . Download mysql connector in Application-server

[illegible]

24 . Configure the Context.xml file present in /opt/apache/conf/context.xml

Vim context.xml file

```
root@ip-10-0-1-204:/opt/apache-tomcat-9.0.121/conf
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements.  See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License.  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- The contents of this file will be loaded for each web application -->
<context>

    <!-- Default set of monitored resources. If one of these changes, the
    <!-- web application will be reloaded. -->
    <watchedResource>WEB-INF/web.xml</watchedResource>
    <watchedResource>WEB-INF/tomcat-web.xml</watchedResource>
    <watchedResource>${catalina.base}/conf/web.xml</watchedResource>

    <!-- Uncomment this to disable session persistence across Tomcat restarts -->
    <!--
    <manager pathname="" />
    -->

    <Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
    maxTotal="500" maxIdle="30" maxWaitMillis="1000"
    username="admin" password="admin12345" driverClassName="com.mysql.jdbc.Driver"
    url="jdbc:mysql://my-database.cvqyci0karn6.ap-south-1.rds.amazonaws.com:3306/studentapp?useUnicode=yes&characterEncoding=utf8"/>
</context>
~
~
"context.xml" 36L, 1752B                                     34,75      A11
```

25 . Configure the nginx.conf Present in jump-server's Nginx Folder . Add Proxy_pass

```
root@ip-10-0-0-200:/etc/nginx
# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile        on;
    tcp_nopush      on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/nginx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen      80;
        listen      [::]:80;
        server_name _;
        root        /usr/share/nginx/html;

        location / {
            proxy_pass http://10.0.1.204:8080;
        }

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;
    }
}

"nginx.conf" 86L, 2392B                                     44,36-43      19%
```

26 . Hit the Jump server's ip address/student on Browser
Application is Running successfully .

Fill The Form .

Student Registration Form

Student Name	Vishal Konde
Student Address	Mumbai
Student Age	24
Student Qualification	mca
Student Percentage	99 %
Year Passed	2026

27 . Data Is Successfully getting Stored in to the Database .

[Register Student](#)

Students List

Student ID	StudentName	Student Addr	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	Vishal Konde	Mumbai	24	mca	99 %	2026	edit	delete

Services Used

- **Amazon EC2** – Hosted Nginx and Apache Tomcat servers.
- **Amazon VPC** – Provided the isolated network environment.
- **Amazon RDS (MySQL)** – Stored student registration records.
- **Nginx** – Used as a reverse proxy for routing user requests.
- **Apache Tomcat** – Hosted and executed the Java web application.
- **Security Groups** – Controlled communication between the web, application, and database tiers.
- **Internet Gateway** – Enabled internet access for the public-facing web tier.

Conclusion

The project successfully deployed a three-tier Java web application on AWS using Nginx, Apache Tomcat, and Amazon RDS MySQL. The architecture separates the web, application, and database layers, while Security Groups control communication between them. Student registration data is successfully processed by the Java application and stored in the RDS database.

